

Introduction

Sustainable finance has become a central topic for investors and regulators. As the world shifts toward a low-carbon economy, a key assumption has emerged in the financial industry: that green companies are not only better for the planet but also safer investments, while brown companies carry higher financial risks. However, does this assumption hold true when we look at the data?

The central question of this report is simple to state but complex to answer: Do Green firms genuinely have lower risk profiles than Brown firms?

To answer this, we cannot simply look at average returns. Financial markets are characterized by fat tails and extreme events, such as crashes and crises, that standard models often ignore. Therefore, our project adopts a rigorous quantitative approach to isolate the true effect of ESG criteria on financial risk.

Our analysis is built on three main pillars. First, we start by upgrading the industry-standard J.P. Morgan RiskMetrics framework. Because financial returns are not Gaussian, we implement a model based on Student-t distributions and EWMA volatility. This allows us to accurately measure tail risk, specifically Value-at-Risk and Expected Shortfall, without underestimating the probability of market crashes.

Second, we construct specific Green and Brown portfolios based on emissions data to compare their behavior. Beyond individual risk, we use Copula-based analysis to understand if these portfolios move together during crises and if holding green assets truly provides diversification when the market crashes. We further test this through historical stress tests and Monte Carlo simulations.

Finally, we recognize that risk is often driven by the general economic environment rather than firm-specific characteristics. To solve this, we use a Kalman filter to extract the financial cycle of different sectors. This allows us to verify if the observed risk differences are due to ESG factors or simply the result of market cycles. Through this comprehensive framework, this report aims to provide a nuanced answer to the debate, distinguishing between market myths and statistical reality regarding the safety of green assets.

I) Risk Metrics methodology

A) Motivation: From RiskMetrics to ESG Risk Profiles

In other words, if I sort firms according to their environmental footprint, do I also obtain portfolios with different market risk characteristics, especially in the tails regarding VaR, ES, and crashes? To explore this, I do not start directly with ESG; instead, I first need a reliable risk model. A large part of the industry still uses the historical J.P. Morgan RiskMetrics framework. RiskMetrics assumes that daily returns are Gaussian and that conditional volatility follows an Exponentially Weighted Moving Average (EWMA).

Formally, the RiskMetrics variance recursion is:

$$\sigma_{t,EWMA}^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94$$

and the 99% Gaussian VaR is:

$$VaR_{0.99,t} = \mu_t + \sigma_{t,EWMA} z_{0.99}, \quad z_{0.99} = 2.33.$$

The framework is transparent and easy to implement, which explains why it is a reference in bank risk management. However, it suffers from a major flaw: financial returns are not Gaussian. Instead, they exhibit fat tails, volatility clustering, and asymmetric losses.

If we wish to make statements asserting that brown portfolios are riskier than green ones, we cannot base our conclusions on a model that underestimates tail risk. Therefore, the first step of this work is to rebuild the RiskMetrics model under more realistic assumptions and choose a model that is well calibrated in the tails.

Only once this is done I apply it to ESG portfolios and ask:

Do green & brown portfolios really have different risk profiles?

Why EWMA, Why GARCH and why student distribution?

To modify RiskMetrics, I change 2 ingredients:

Volatility model and Distribution of innovations (errors)

Volatility: EWMA or GARCH?

RiskMetrics uses EWMA:

$$\sigma_{t,EWMA}^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2, \quad \lambda = 0.94$$

Pro: easy to compute, reacts quickly to shocks (t-1 to t)

Cons: No “long-memory” structure

GARCH(1,1):

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2.$$

This model is widely used in banks for stressed VaR and internal models because it captures persistent volatility regimes.

1) Distribution: Gaussian or Student-t

Second part is the conditional distribution of standardized innovations. RiskMetrics assumes they are Gaussian. This is not consistent with the empirical reality of asset returns. To fix this, I made the assumption of a student-t distribution:

$$z_t = \frac{r_t}{\sigma_t} \sim t_v(0,1)$$

The key advantage lies in the presence of the degrees of freedom parameter v . The distribution converges to a Gaussian one as v increases, whereas when v is small, the tails become very heavy, which allows the model to adapt to the actual behavior of market returns. Combining these ingredients regarding volatility models and distributions yields three candidates: the original RiskMetrics using EWMA and Gaussian assumptions, a RiskMetrics-t framework where EWMA is fitted with Student likelihood, and a GARCH(1,1) model coupled with a Student-t distribution.

B) Methodology and Model Selection

We first estimate EWMA & GARCH volatilities on the panel and compare them to realized volatility.

The idea is simple: If a volatility model is good, it should track well the realized volatility. The idea is simple: If a volatility model is good, it should track well the realized volatility.

	vol_std	sigma_ewma	sigma_garch
vol_std	1.000000	0.790282	0.779776
sigma_ewma	0.790282	1.000000	0.931373
sigma_garch	0.779776	0.931373	1.000000

EWMA outperforms GARCH on this dataset (I could be the tuning of GARCH the problem as well).

Standardized returns

$$z_t = \frac{r_t}{\sigma_t}$$

Under a correctly specified model, these residuals should be approximately i.i.d (and student-t).

Fitting the Student-t Distribution

$$z_t = \frac{r_t}{\sigma_t} \sim t_v(0,1)$$

We fit a Student-t distribution on z_t .

VaR modelling & backtesting

$$VaR_{\alpha,t} = \sigma_t q_\alpha$$

This is the formula for a parametric VaR (Value-at-Risk)

Now the key selection between 3 model is how well each one matches the historical VaR level based on data. We focus on 99% VaR and compute the exception rate (% of days where the realized loss exceeds the model's VaR).

Model	VaR99 exception rate
EWMA + Gaussian (RiskMetrics)	1.11%
EWMA + Student-t	1.01%
GARCH(1,1) + Student-t	0.35%

The results at the 1% target reveal that RiskMetrics underestimates tail risks with too many exceptions, while the GARCH-Student-t model proves too conservative with very few. In contrast, the EWMA combined with a Student-t distribution is almost perfectly calibrated at 1.01%, aligning closely with the 1% objective. From a risk management perspective, this accuracy is ideal. We therefore select the EWMA-Student-t model because it maintains the simplicity of the original J.P. Morgan framework while effectively correcting for fat tails and ensuring robust calibration. With this foundation set, we move to ESG feature construction, shifting focus from emissions to portfolios.

ESG feature construction: from emissions to portfolios

The idea is to build an emissions-based ESG indicator and then construct brown & green portfolios.

I start from Scope 1,2 & 3 each normalized by Sales. For each scope $i \in 1, 2, 3$, I define a simply binary:

$$\text{binary}_i = \begin{cases} 1 & \text{if emissions/sales} > 0.5, \\ 0 & \text{otherwise.} \end{cases}$$

We classify firms as brown if the score is zero, neutral if it equals one, and green if it is greater than 1.5. This represents a simple approach with a clear economic interpretation in which firms with consistently high emissions across scopes are classified as brown and vice versa.

Simple approach with a clear economic interpretation. Firms with consistently high emissions across scopes are classified as brown and vice versa.

ESG statistics backtest

Green Firm			Neutral Firm			Brown Firm					
	Scope1_Sales	Scope2_Sales	Scope3_Sales		Scope1_Sales	Scope2_Sales	Scope3_Sales		Scope1_Sales	Scope2_Sales	Scope3_Sales
count	74004.000000	74004.000000	74004.000000	count	116912.000000	116912.000000	116912.000000	count	273578.000000	273578.000000	273578.000000
mean	0.755104	1.260756	678.764370	mean	11.381273	10.392550	967.897189	mean	216.880387	40.598892	1878.997681
std	1.369546	2.926114	2140.699608	std	42.388130	23.163934	2444.255735	std	659.222852	80.863426	2953.914936
min	0.030000	0.210000	6.120000	min	0.260000	0.210000	14.380000	min	1.000000	1.010000	103.590000
25%	0.280000	0.440000	64.670000	25%	1.240000	0.810000	53.850000	25%	6.380000	7.350000	329.730000
50%	0.400000	0.480000	84.170000	50%	2.300000	2.580000	113.620000	50%	13.254867	15.330000	929.140000
75%	0.650000	1.940000	297.530000	75%	4.290000	9.220000	332.680000	75%	68.190000	27.520000	2012.600000
max	6.360000	30.560000	9551.749576	max	234.550000	121.590000	12750.413215	max	4180.810000	508.320000	25349.000000

The observed differences are significant, confirming that the green score serves as a valid proxy for decarbonization. The question now is whether these substantial variations in emissions translate into distinct risk profiles. To investigate this, I construct two portfolios, Green and Brown, based on the mean returns of all firms within each category. This section addresses the central research question of whether green and brown portfolios display different risk characteristics when analyzed through a well-calibrated risk model. We begin by analyzing tail thickness through the Student-t degree of freedom, specifically looking at the estimated parameters derived from the Student Maximum Likelihood Estimation for each portfolio.

Tail Thickness: Student-t Degree of Freedom

We first look at the estimated degrees of freedom from the Student MLE on each portfolio:

Portfolio	(v)
Brown	9.51
Green	12.51

A lower v mean fatter tails. The brown portfolio therefore has clearly heavier tails than the green portfolio. This already suggests that brown assets are more prone to extreme movements.

VaR99 Backtests by ESG category

We compute 99% VaR (EWMA Student-t) on both portfolios.

Hit Ratio remains fine after the data manipulation (from 500K rows to 5k rows) which logically conduce to deterioration of the previous VaR.

```
Hit ratio VaR99 brown: 1.358695652173913 %  
Hit ratio VaR99 green: 1.4328063241106719 %
```

```
count      4048.000000  
mean        0.029348  
std         0.014317  
min         0.000500  
25%         0.019484  
50%         0.025927  
75%         0.034888  
max         0.113147  
Name: VaR_ret_99_brown, dtype: float64
```

```
count      4048.000000  
mean        0.026705  
std         0.011850  
min         0.007576  
25%         0.018306  
50%         0.024053  
75%         0.031313  
max         0.096166  
Name: VaR_ret_99_green, dtype: float64
```

Both are close to the 1% target, but here what matters is mainly the level of VaR and the overall shape of distribution. We constate that the mean of Brown VaR is higher than Green own, same for standard deviation.

Expected Shortfall (ES): Average loss in the tail

This metric tells us how deep the average loss is once we are in the tail.

$$ES_{\alpha,t} = \sigma_t E(z_t | z_t < q_\alpha)$$

$$ES_\alpha = \sigma \cdot \frac{\nu + q_p^2}{(\nu - 1)p} f_\nu(q_p)$$

For 99% ES under our model, we obtain:

Brown portfolio:

- Empirical $ES_{99} \approx 3.3048\%$
- Model $ES_{99} \approx 3.1951\%$
- Ratio $ES_{\text{empirical}} / ES_{\text{model}} \approx 1.043$

Green portfolio:

- Empirical $ES_{99} \approx 2.7808\%$
- Model $ES_{99} \approx 2.5644\%$
- Ratio ≈ 1.077

We point out some things:

Level of risk: ES is clearly higher for brown than green. So in extremes scenarios, the average loss in the tail is larger for brown than green.

Model calibration: ratios are close to 1 for both, which is what we want (under conditions of small dataset).

Pulling together evidences:

Emissions: brown firms have massively higher Scope 1/2/3 emissions → clear environmental difference.

Tail shape: the brown portfolio has a lower $v(9.51 \text{ vs } 12.51)$, meaning **heavier tails**.

VaR: brown VaR levels are higher, consistent with a riskier tail.

ES: brown ES is significantly higher than green ES in the 99% tail region.

We can say: Yes, ESG criteria (on emissions) do translate into different risk profiles at the marginal level: the Green portfolio is safer than the Brown portfolio.

However, from a systemic risk perspective, it is not enough to know that brown assets have heavier tails individually. We need to know if:

Green & Brown portfolios crash together?

In crisis does holding green assets provide diversification?

II) Copula-Based Dependence Analysis of Brown and Green Portfolios

This section presents the copula-based analysis for the brown and green portfolios. It is designed as one step within the overall paper, with the goal of understanding how ESG criteria may (or may not) translate into different risk profiles, especially in extreme scenarios.

1. Motivation: Why a Copula Step?

In the first part of the paper, I analyse the brown and green portfolios separately using descriptive statistics and standard risk measures such as Value-at-Risk (VaR). A simple comparison already suggests that the brown portfolio is slightly riskier than the green one: its VaR is higher for most confidence levels, and its return distribution is somewhat more volatile.

However, this marginal analysis misses a crucial question: do brown and green portfolios move together in normal and extreme conditions, and does the ESG profile really translate into a different joint risk profile? From a risk management perspective, what matters is not only whether the green portfolio is less risky on its own, but also whether it diversifies the risk of the brown portfolio, especially in crisis situations. If both portfolios crash at the same time, then ESG exposure does not provide much protection in systemic stress episodes.

This motivates the copula part of the paper. The objective of this section is to analyse how the brown and green portfolios move together in normal times and in the extremes, to check whether joint crashes are frequent (tail dependence), and to see whether the dependence structure is consistent with a symmetric Student-t copula or whether the data point towards asymmetric dependence. The final goal is to connect this dependence analysis back to the ESG question: does the introduction of ESG criteria genuinely change the risk profile, or do green and brown portfolios remain strongly linked in bad states of the world?

2. Why Copulas, and Why a Student-t Copula?

Classical correlation (Pearson's (ρ)) is not sufficient to answer these questions. It captures only linear co-movements, is sensitive to scaling, and does not distinguish between normal and extreme states. It also cannot tell us whether joint tail events (co-crashes or co-booms) are more likely than under independence.

Copulas provide a natural way to separate the marginal behaviour of each portfolio (volatility, skewness, fat tails) from the dependence structure (how the two portfolios move together over the entire distribution and in the tails). In this paper, I focus on a Student-t copula for three reasons:

The whole modelling framework of the paper is built around Student-t distributions for returns/losses, which naturally capture fat tails.

The t-copula generalises the Gaussian copula by allowing tail dependence, which is critical for joint crash risk.

It remains relatively simple and interpretable: it is parametrised by a correlation parameter (ρ) and a degrees-of-freedom parameter (ν_{cop}), directly linked to the strength of dependence and the thickness of the tails.

The Student-t copula is therefore a coherent choice with the rest of the modelling: margins and copula share the same “Student-t logic”, making the structure easy to interpret as a whole.

3. Methodology

3.1 Step 1 – Modelling the Margins with AR–GARCH–t

The first step is to cleanly model the marginal distributions of the brown and green portfolio returns. In practice, large financial institutions rarely work with raw returns; they typically filter out predictable dynamics, such as mean reversion and autocorrelation, and heteroskedasticity (changing volatility), and then model the innovations with a heavy-tailed distribution.

To mimic this kind of high-end risk engine, I estimate for each portfolio $k \in \{\text{brown, green}\}$ an AR(1)–GARCH(1,1) model with Student-t innovations. For portfolio k , the return decomposition is:

$$r_t^{(k)} = \mu_t^{(k)} + \varepsilon_t^{(k)}$$

The conditional mean follows an AR(1) specification:

$$\mu_t^{(k)} = \alpha_0^{(k)} + \alpha_1^{(k)} r_{t-1}^{(k)}$$

The conditional variance is modelled as a GARCH(1,1):

$$\varepsilon_t^{(k)} = \sigma_t^{(k)} z_t^{(k)}$$

$$(\sigma_t^{(k)})^2 = \omega^{(k)} + \alpha^{(k)} (\varepsilon_{t-1}^{(k)})^2 + \beta^{(k)} (\sigma_{t-1}^{(k)})^2$$

The innovations are assumed to follow a Student-t distribution with ν_k degrees of freedom:

$$z_t^{(k)} \sim t_{\nu_k}$$

The standardized residuals are then defined as:

$$z_t^{(k)} = \frac{\varepsilon_t^{(k)}}{\sigma_t^{(k)}}$$

This captures volatility clustering (via GARCH), autocorrelation in returns (via AR), and fat tails (via the Student-t innovations). The result of this step is a sequence of conditional volatilities $(\sigma_t^{(k)})$ and a sequence of standardized residuals $(z_t^{(k)})$. These residuals are the innovations used to build the copula.

3.2 Step 2 – Full Parametric PIT: Transforming to Uniforms

The second step is to transform the residuals into uniform variables on $[0,1]$, which is what copulas work with. Instead of using a semi-parametric rank-based transform, I deliberately keep a fully parametric approach to remain consistent with the overall Student-t framework of the paper.

Under the AR–GARCH–t model, $(z_t^{(k)})$ is approximately Student-t with (ν_k) degrees of freedom. Let (F_{t,ν_k}) denote the CDF of the Student-t distribution with (ν_k) degrees of freedom. The Probability Integral Transform (PIT) is then defined as:

$$U_t^{(k)} = F_{t,\nu_k}(z_t^{(k)})$$

For the brown and green portfolios, this gives:

$$U_t^{(\text{brown})} = F_{t,\nu_{\text{brown}}}(z_t^{(\text{brown})})$$

$$U_t^{(\text{green})} = F_{t,\nu_{\text{green}}}(z_t^{(\text{green})})$$

Practically, for each time t , I compute these PIT values and clip values extremely close to 0 or 1 to avoid numerical issues. The result is a sample $(U_t^{(\text{brown})}, U_t^{(\text{green})})$ that represents the joint positions of the portfolios in their own Student-t distributions, after controlling for dynamics and volatility.

3.3 Step 3 – Global Dependence on the Copula Level

Before fitting a copula, it is informative to look at global dependence measures on the uniforms: Spearman's rank correlation and Kendall's tau. Both are computed directly on $(U^{(\text{brown})}, U^{(\text{green})})$, and are therefore purely copula-based (invariant to marginal transformations).

Let (R_i) be the rank of $(U_i^{(\text{brown})})$ and (S_i) the rank of $(U_i^{(\text{green})})$, for $i = 1, \dots, n$

Spearman's rho (simple form without ties) is:

$$\rho_s = 1 - \frac{6 \sum_{i=1}^n (R_i - S_i)^2}{n(n^2 - 1)}$$

More generally (as a Pearson correlation on ranks):

$$\rho_s = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2} \sqrt{\sum_{i=1}^n (S_i - \bar{S})^2}}$$

Kendall's tau is based on concordant and discordant pairs. Let (N_C) be the number of concordant pairs and (N_D) the number of discordant pairs among the $(n(n-1)/2)$ possible pairs. Then the sample Kendall's tau is:

$$\tau = \frac{N_C - N_D}{\frac{1}{2}n(n-1)}$$

For elliptical copulas such as the Gaussian or t-copula, there is a direct link between the copula correlation parameter (ρ) and these rank-based measures:

$$\tau = \frac{2}{\pi} \arcsin(\rho)$$

$$\rho_s = \frac{6}{\pi} \arcsin\left(\frac{\rho}{2}\right)$$

Empirically, I find that Spearman's rho from data is high (around 0.83) and Kendall's tau from data is also high (around 0.65), with p-values essentially zero. This means that even after filtering out marginal effects, the brown and green portfolios exhibit a very strong monotonic dependence: when the brown portfolio moves to a high quantile of its conditional distribution, the green portfolio tends to move to a high quantile as well. ESG criteria do not, by themselves, break the global dependence between the two portfolios.

3.4 Step 4 – Fitting the Student-t Copula

Next, I fit a bivariate Student-t copula to the uniforms $(U^{(\text{brown})}, U^{(\text{green})})$.

The t-copula is characterised by a correlation parameter (ρ), which controls the overall strength of dependence, and a degrees-of-freedom parameter (ν_{cop}), which controls the thickness of the copula's tails.

I estimate (ρ, ν_{cop}) by maximum likelihood using the known density of the t-copula. The estimates are: $(\hat{\rho} \approx 0.88)$ and $(\hat{\nu}_{\text{cop}} \approx 5)$. Interpreting these numbers, the copula correlation confirms very strong dependence between the portfolios once margins are controlled for, and the moderate (ν_{cop}) indicates heavy tails in the dependence structure itself.

From the fitted t-copula, I can also derive the theoretical Kendall's tau and Spearman's rho implied by the model using the formulas above. These are slightly higher than the empirical ones but of the same order of magnitude, which suggests that the t-copula reproduces the overall strength of dependence reasonably well.

3.5 Step 5 – Tail Dependence: Lower and Upper Tails

To understand how the portfolios behave in extreme scenarios, I focus on tail dependence coefficients. Let $(U_1, U_2) = (U^{(\text{brown})}, U^{(\text{green})})$ be the uniforms with copula (C) . The lower-tail dependence (λ_-) measures the probability that the

green portfolio is also extremely low when the brown portfolio is in its lowest quantiles. The upper-tail dependence (λ_+) measures the probability that the green portfolio is also extremely high when the brown portfolio is in its highest quantiles.

The theoretical definitions are:

$$\lambda_- = \lim_{u \rightarrow 0^+} P(U_2 \leq u \mid U_1 \leq u) = \lim_{u \rightarrow 0^+} \frac{C(u, u)}{u}$$

$$\lambda_+ = \lim_{u \rightarrow 1^-} P(U_2 > u \mid U_1 > u) = \lim_{u \rightarrow 1^-} \frac{1 - 2u + C(u, u)}{1 - u}$$

Empirically, I approximate these limits by evaluating, for small or large quantiles q :

$$\widehat{\lambda}_-(q) \approx \frac{\widehat{P}(U_1 \leq q, U_2 \leq q)}{q}$$

$$\widehat{\lambda}_+(q) \approx \frac{\widehat{P}(U_1 > q, U_2 > q)}{1 - q}$$

For a bivariate Student-t copula with correlation parameter (ρ) and degrees of freedom (v_{cop}), the theoretical tail dependence is symmetric and given by:

$$\lambda_+ = \lambda_- = 2 T_{v_{\text{cop}}+1} \left(- \sqrt{\frac{(v_{\text{cop}} + 1)(1 - \rho)}{1 + \rho}} \right)$$

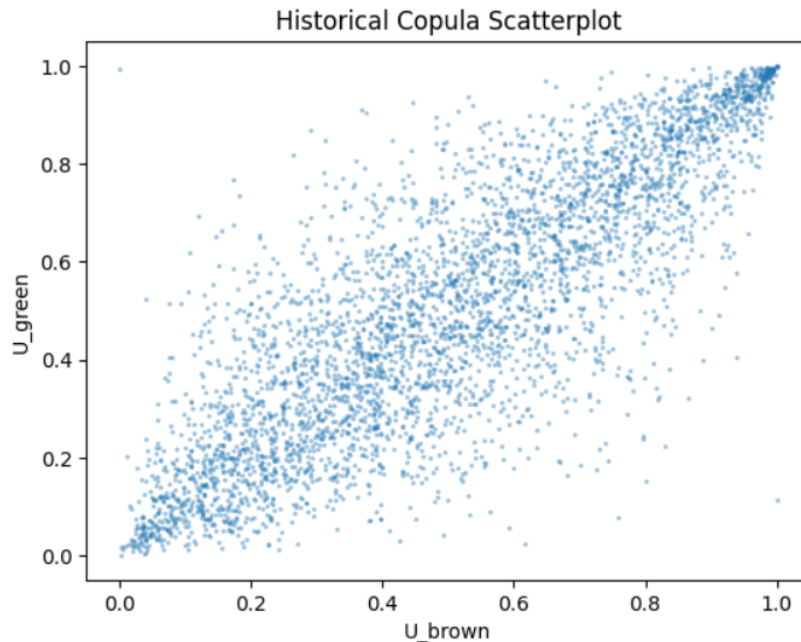
where ($T_{v_{\text{cop}}+1}$) is the CDF of a univariate Student-t distribution with ($v_{\text{cop}} + 1$) degrees of freedom.

Empirically, the results are asymmetric. For the lower tail, as q decreases (for example $q = 0.05, 0.02, 0.01$), the empirical ($\widehat{\lambda}_-(q)$) falls towards zero. This indicates that joint extreme events in the lower tail are rare: when one portfolio is in its worst quantiles, the other is not systematically in an equally bad state. For the upper tail, for large q (for example $q = 0.90, 0.95, 0.98$), ($\widehat{\lambda}_+(q)$) remains around 0.5–0.55. This means that, in extreme upper-tail events, there is roughly a 50% probability that both portfolios are simultaneously in their own extremes.

In other words, the portfolios show a strong tendency to be extreme together on one side of the distribution, while they are much less frequently extreme together on the other side.

4. Symmetric t-Copula versus Empirical Asymmetry

A symmetric t-copula imposes the same tail dependence in both tails: ($\lambda_+ = \lambda_-$). For the estimated parameters ($\widehat{\rho} \approx 0.88$) and ($\widehat{v}_{\text{cop}} \approx 5$), the theoretical tail dependence of the t-copula is around 0.55 for both tails.



However, the empirical results ($(\lambda_+ \approx 0.5) - 0.55$ and $(\lambda_- \approx 0)$) and the copula scatterplot show a clear asymmetry: dependence is strong in the upper tail but very limited in the lower tail. The symmetric t-copula accurately reproduces the overall dependence and part of the extreme co-movements, but it overestimates lower-tail dependence.

This suggests that a copula with dominant upper-tail dependence, such as the Gumbel copula, or a more general asymmetric copula model, would be better suited to faithfully capture the dependence structure observed between the brown and green portfolios.

5. Implications for ESG and Risk Profiles

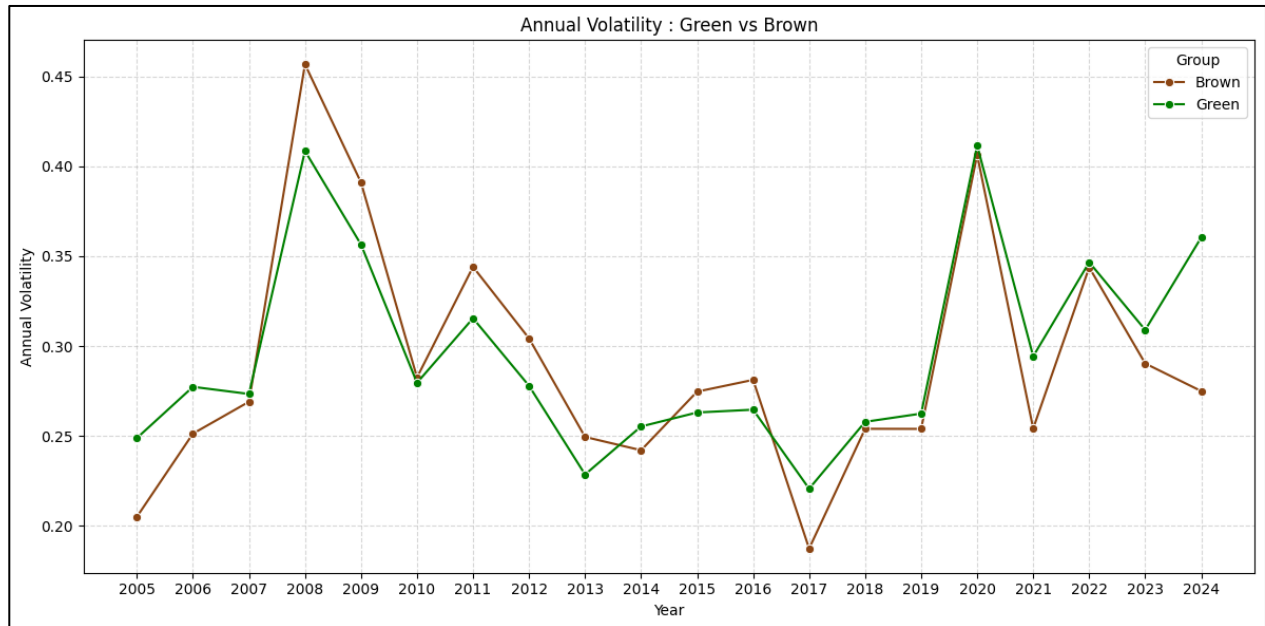
Putting these results together, the copula analysis shows that ESG criteria do not fully decouple the green portfolio from brown risk. From a marginal perspective, the brown portfolio appears slightly riskier than the green portfolio: its VaR is consistently higher. From a joint perspective, the copula analysis reveals that brown and green remain strongly linked, especially in extreme conditions on one side of the distribution.

In normal conditions, the portfolios share a high level of dependence, as shown by the high Spearman and Kendall coefficients and the large copula correlation parameter. In extreme scenarios, upper-tail dependence is strong, meaning that severe shocks tend to affect both portfolios at the same time. However, lower-tail dependence is very weak in the data, and the symmetric t-copula fails to capture this asymmetry.

Overall, the copula analysis suggests that ESG criteria do change the shape of the risk profile (slightly lower VaR and different tail behaviour), but they do not fully isolate the green portfolio from systemic risk. The two portfolios still share a substantial amount of joint extreme risk, and the dependence structure is more complex and asymmetric than what a simple symmetric t-copula can express.

III) Who is more Stressed: Green or Brown ?

The graph shows that the annual volatility of Green and Brown firms generally follows the same trend. However, during major market shocks, such as the Subprime crisis or the Covid pandemic, we observe that Brown firms tend to display higher volatility than Green firms. To confirm this, we conducted stress tests using Expected Shortfall and Value-at-Risk (VaR) to analyze how volatility behaves and to verify the hypothesis that Brown companies are riskier than Green ones.



To verify our assumption, we conduct various stress tests to determine if Green firms generally exhibit lower financial risk than Brown firms. First, we calculate the daily Value-at-Risk (VaR) at a 99% confidence level and the Expected Shortfall (ES) at 97.5% for the year 2008.

The VaR quantify potential maximum loss under 99% probability while ES quantify the average loss in extreme scenarios bigger than the VaR. We use two specific weighting methods for this analysis. One is based on Scope 1 emissions relative to sales to place more emphasis on the most polluting firms, while the other is based on the logarithm of capitalization to account for the influence of firm size.

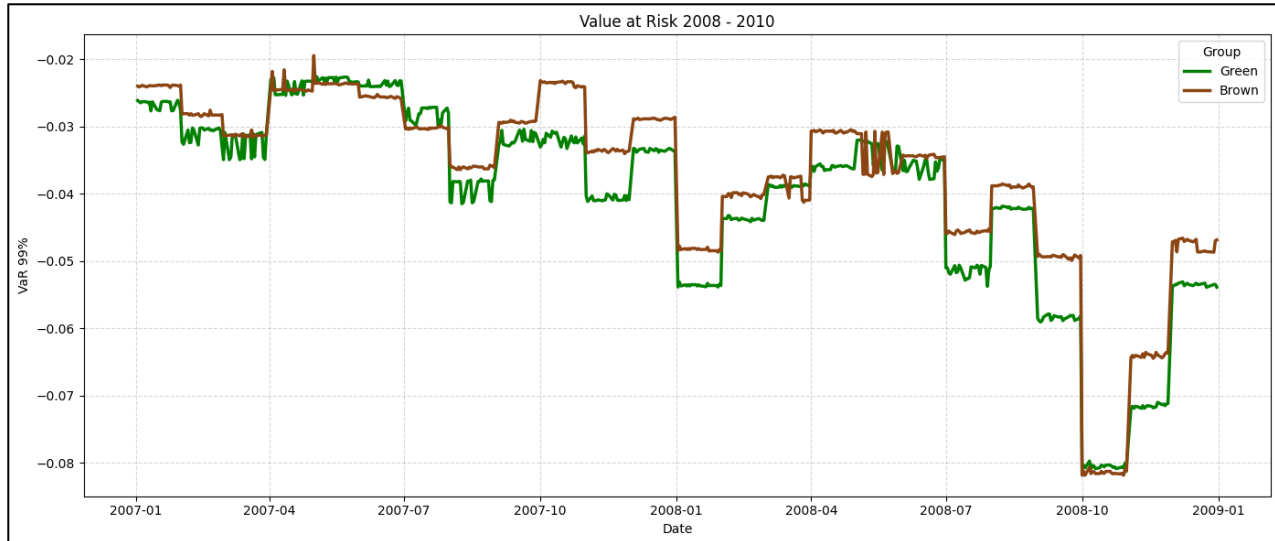
Second, we apply these stress tests to different economic sectors using the same weighting methods to see if the composition of the sector drives the risk measure.

Third, we run a Monte Carlo simulation by applying artificial shocks of +20% and -20% to the returns of Brown firms to observe how risk measures change under these extreme conditions. This approach allows us to analyze risk from different angles, covering firm-level historical data, sectoral analysis, and simulated theoretical shocks.

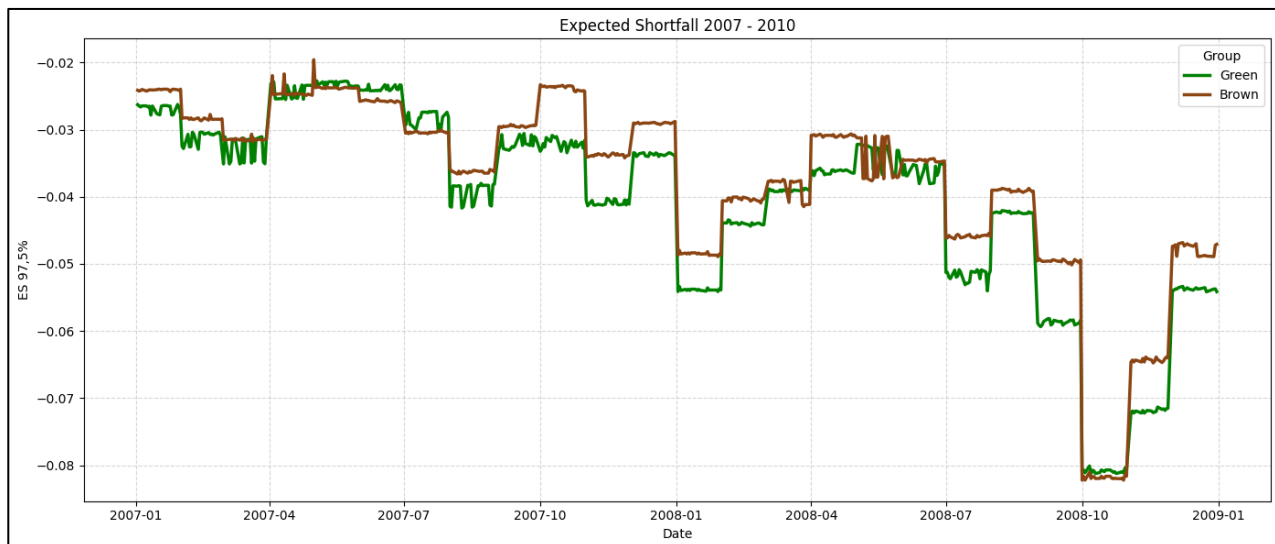
A. Stress Test on green and brown cluster: Subprime Period

To see if the risk is different between green and brown firms, we calculated daily Value-at-Risk (VaR) at 99% and Expected Shortfall (ES) at 97.5% for the year 2008 to put a Stress on the volatility and see if green are less risky than brown.

The results tell us that on average during moderate volatility, Green firms actually showed slightly higher risk than Brown firms during this period. The average daily VaR for Green firms was -4.11%, compared to -3.87% for Brown firms. We saw a similar trend with the Expected Shortfall, where Green firms had an average daily loss of -4.13%, while Brown firms averaged were -3.89%. This suggests that on a typical bad day in 2008, Green firms were slightly more volatile.

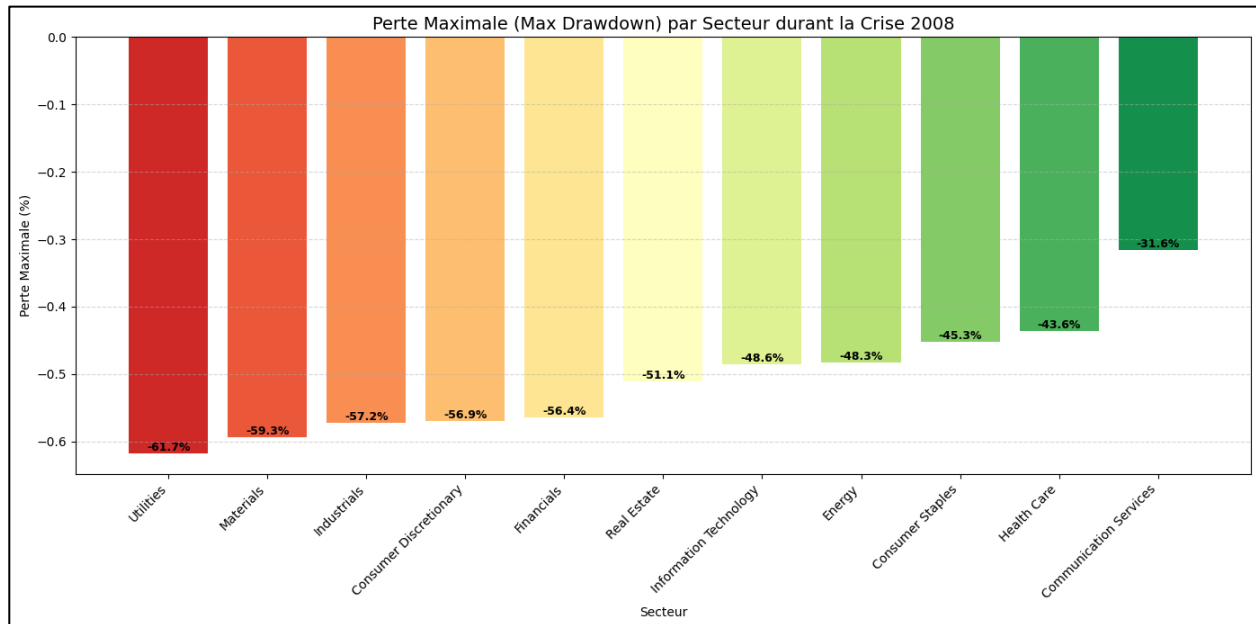


However, the picture changes when we look at the absolute worst days of the crisis, known as the extreme tails. In these extreme scenarios, Brown firms suffered larger losses than Green firms. The worst observed VaR for Brown firms was -8.19% and the worst Expected Shortfall was -8.22%. In comparison, Green firms performed slightly better with -8.09% and -8.12% respectively. While a difference of 0.10% might seem small, it is critical for a bank. Under regulations like the Basel Committee rules, even a tiny increase in extreme potential losses can drastically increase the amount of capital a bank must hold in reserve to remain safe.

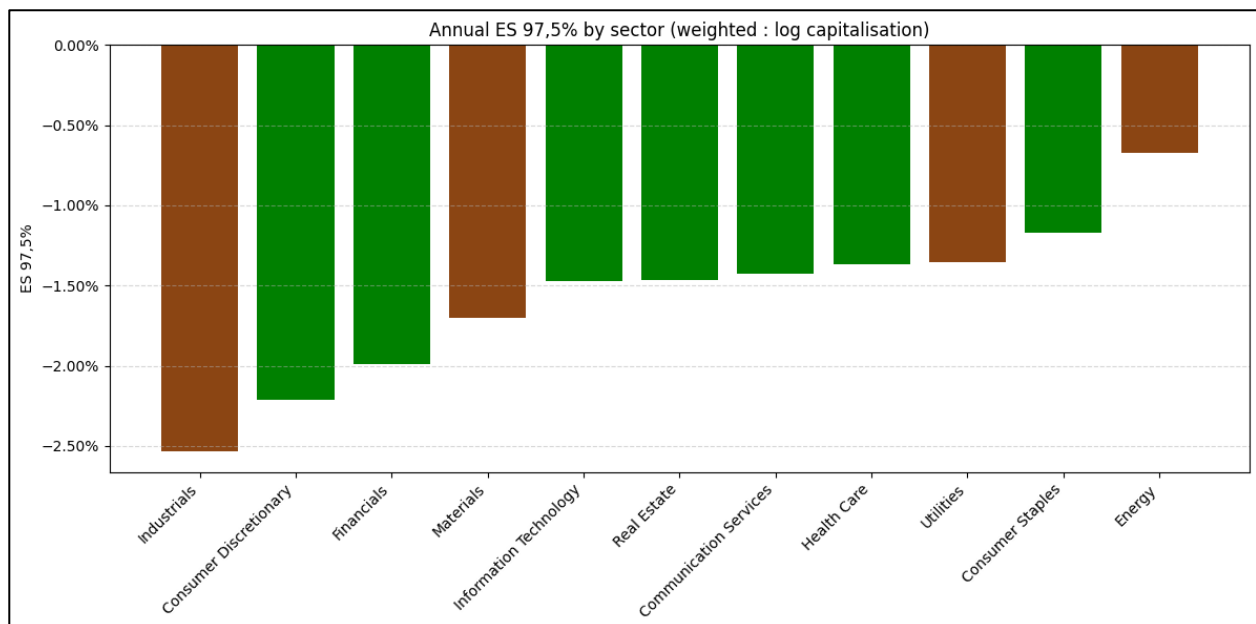


B. Stress Test applied on sector during 2008

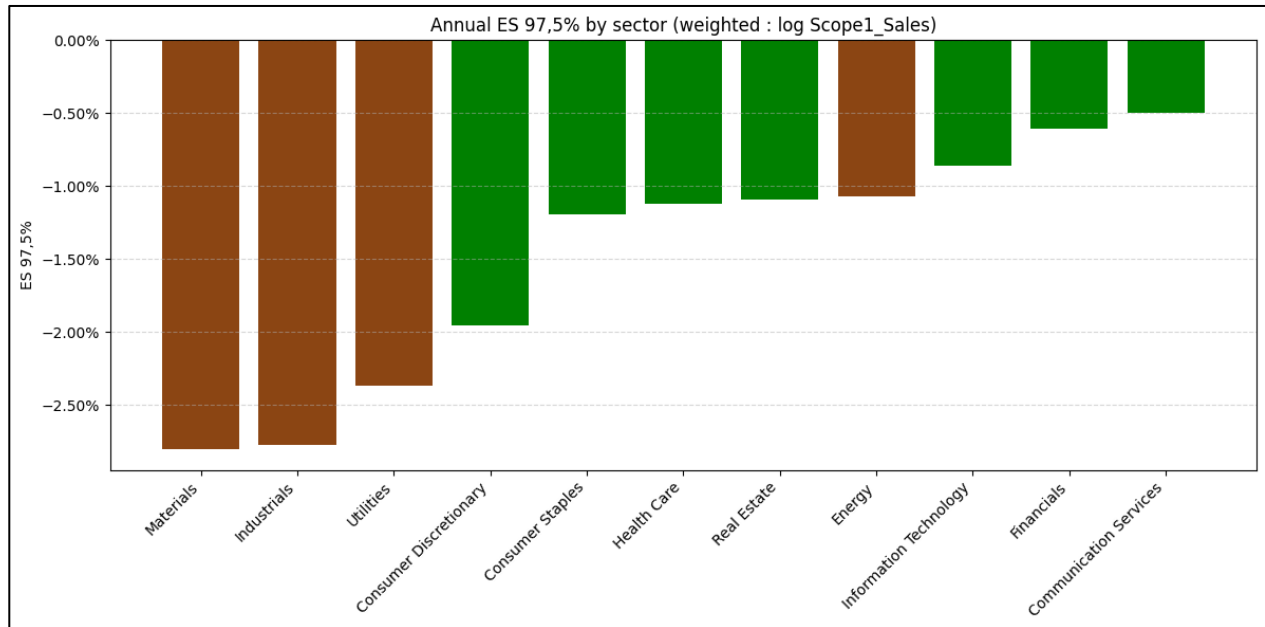
To better understand where the risks lie within the economy, we compared the results using two different weighting methods. The first method uses market capitalization, giving more importance to the largest firms to reflect their financial impact on the market. The second method uses Scope 1 emissions relative to sales, which emphasizes the most polluting firms to highlight environmental exposure. By comparing these two approaches, we can see how firm characteristics influence risk levels and identify which sectors are truly driving volatility depending on whether we look at financial size or environmental footprint.



When we weight the sectors by firm size, the riskiest areas are Industrials and Consumer Discretionary. These sectors showed average daily losses (VaR) of -1.3% and -1.15%, with the worst extreme losses reaching -2.35% and -1.88% respectively. In this view, sectors like Energy, Consumer Staples, and Financials appear to have lower risk. This confirms that under a traditional financial view, the largest firms dominate the risk profile. This means a sector can appear risky simply because it contains very large companies, even if they are not the most polluting.



However, when we weight the sectors by pollution levels, the risk landscape changes. The Materials and Industrials sectors clearly stand out as the riskiest, with average potential losses of -1.64% and -1.58%. The Utilities sector also appears much riskier than it did under the size-based weighting, which reflects the high environmental intensity of these industries. In contrast, low-emission sectors like Financials, Communication Services, and Information Technology became largely less risky under this weighting method.



Comparing these two approaches demonstrates that the choice of weighting is crucial. Capitalization weighting highlights sectors dominated by large firms where financial scale drives the risk. On the other hand, pollution weighting exposes the specific risks of high-emission sectors like Materials, Industrials, and Utilities. Interestingly, some sectors like Energy or Consumer Staples appear much less risky under pollution weighting than under size weighting, proving that financial size and environmental impact do not always align. This comparison is vital for investors and risk managers, as it allows them to adjust the assets in their portfolios to specifically reduce either financial or environmental risk.

C. Shock Test on Polluters returns: Monte Carlo Simulation

- *Monte Carlo simulation with pollution weighted*

To assess the financial impact of climate-related shocks on firms with different environmental profiles, we employed a Monte Carlo simulation. This method is used in risk management because it allows us to model the behavior of complex systems under uncertainty by generating many random scenarios. In this specific case, we simulated the daily returns of firms, taking into account their classification as Green or Brown based on their Scope 1 emissions. This approach is particularly useful for capturing extreme events and tail risks that are not always visible in simple historical analysis.

Our dataset covered the historically volatile period from 2007 to 2008. We classified firms as Brown if they had higher emissions median and Green otherwise. Crucially, each firm's impact on the overall volatility was weighted on its Scope 1 by sales emissions. This reflects the logic that more polluting firms would contribute more significantly to potential losses in a climate stress scenario.

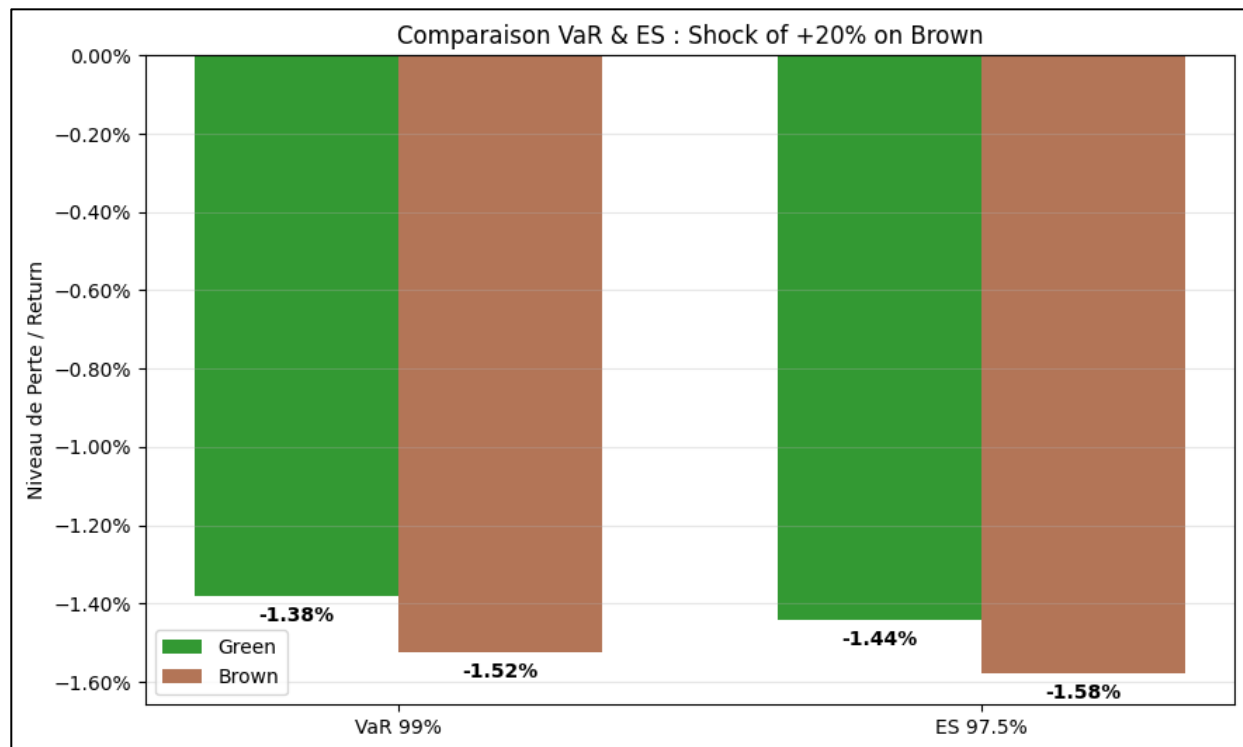
We introduced two specific stress scenarios to test the resilience of these firms. The first was a positive shock of 20% on the returns of Brown firms, simulating an unexpected performance boost for high emitters. The second was a negative shock of 20%, representing a severe crash for polluting companies.

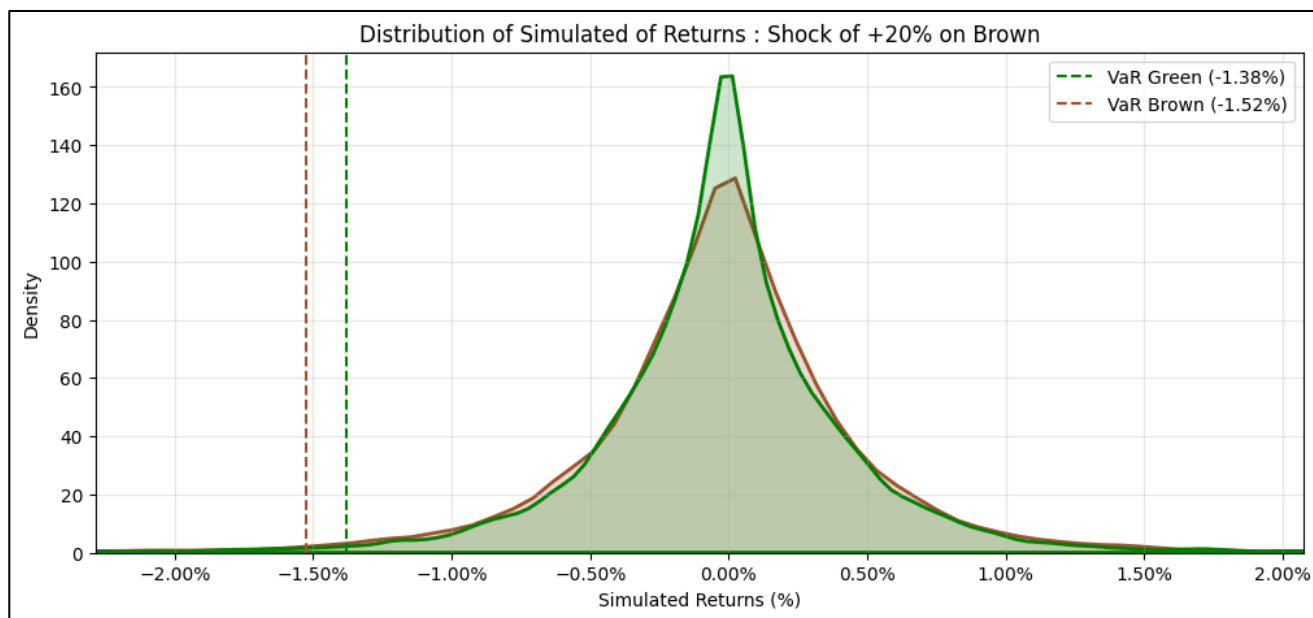
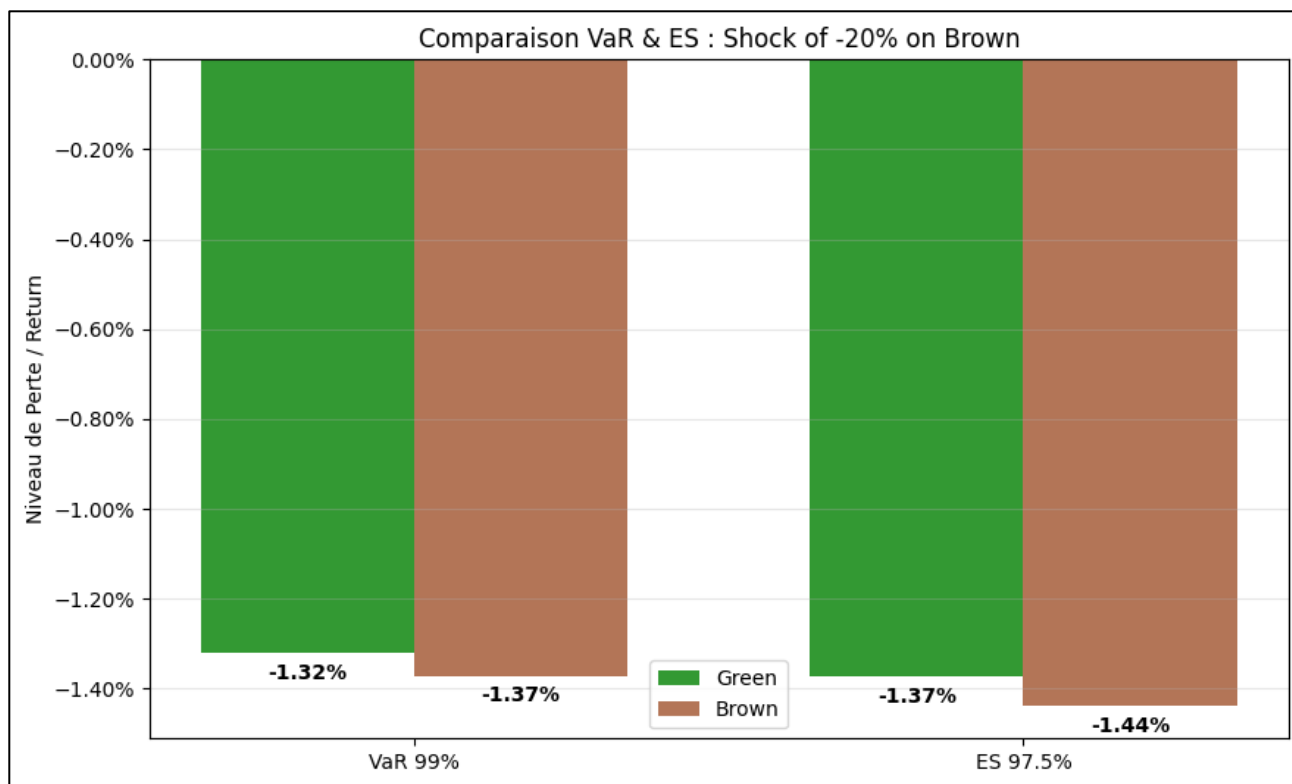
We ran this simulation two million times, sampling daily returns for Green and Brown firms separately under these conditions. For each iteration, we calculated the Value at Risk at 99% and the Expected Shortfall at 97.5% to measure both the threshold of extreme losses and the average loss in the worst-case scenarios.

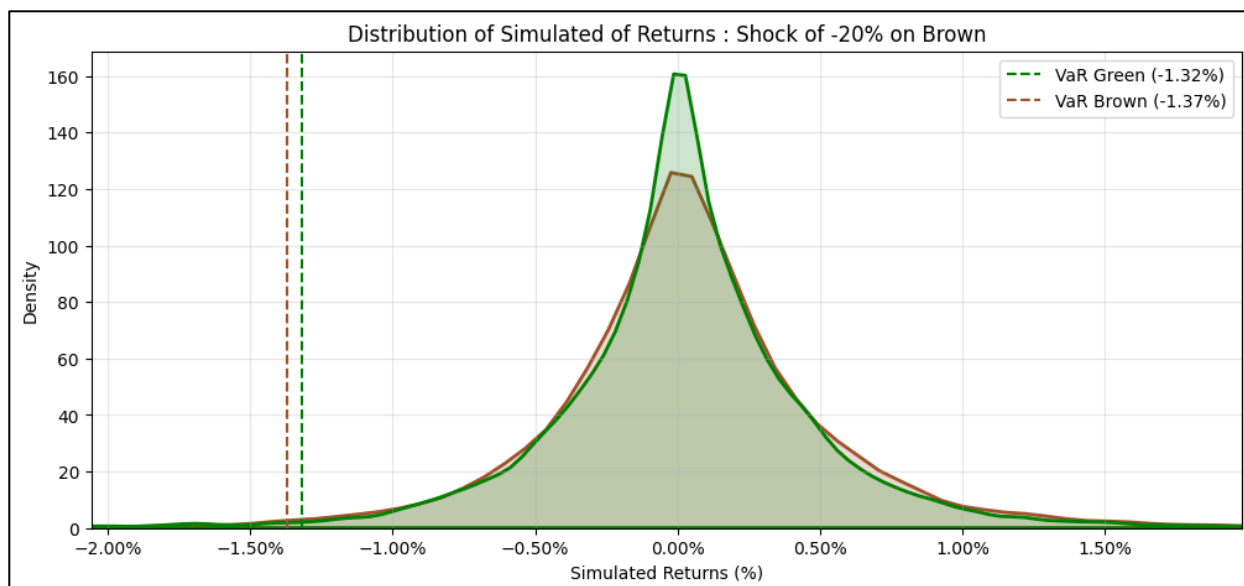
Results of daily risk Metrics simulated:

Shock	Group	VaR 99%	ES 97.5%
-20% on brown returns	Green	-1.32%	-1.37%
-20% on brown returns	Brown	-1.37%	-1.44%
+20% on brown returns	Green	-1.38%	-1.44%
+20% on brown returns	Brown	-1.52%	-1.57%

The results demonstrate that Brown firms carry higher tail risk compared to Green firms, even when we simulate a positive shock. Even when the returns of polluting firms improve by 20%, the risk metrics for Brown firms still predict larger expected losses in extreme scenarios. In contrast, Green firms, which have lower emissions and hold less weight in the stressed portfolio, exhibit smaller expected losses. This indicates that environmental exposure significantly influences tail risk: highly polluting firms represent the main drivers of extreme losses, whereas low-emission firms contribute much less to these severe outcomes. This comparison also highlights that stress scenarios based on emissions are more sensitive than simple market-weighted shocks, emphasizing the importance of including ESG factors in risk analysis.







The results of the Monte Carlo simulation, weighted by Scope 1 emissions by sales, confirm that Brown firms present higher tail risk than Green firms across both stress scenarios.

- In the first scenario (Shock of -20%): Brown firms present a Value-at-Risk (VaR 99%) of -1.3712% compared to -1.3194% for Green firms. Similarly, the Expected Shortfall (ES) indicates that extreme losses are deeper for the Brown cluster (-1.4391%) than for the Green cluster (-1.3735%).
- In the second scenario (Shock of +20%): The gap widens significantly. Brown firms show a VaR of -1.5241%, indicating a much higher potential loss than Green firms (-1.3825%). The Expected Shortfall follows the same trend, with Brown firms reaching -1.5793%, confirming that highly polluting firms are the primary drivers of extreme downside risk when volatility increases.

D. Is being brown bring more volatility?

To isolate the impact of environmental performance on risk, we employ an Ordinary Least Squares (OLS) regression model. The dependent variable is the volatility.

The key independent variables of interest are defined as binary indicators:

- **Is_High_ESG**: A dummy variable equal to 1 if the firm's ESG score is strictly greater than 70, and 0 otherwise.
- **Is_Brown**: A dummy variable equal to 1 if the firm's Scope 1 emissions-to-sales ratio is above the sample median, and 0 otherwise.

We control for fundamental financial characteristics using the following metrics:

- **LeverageRatio1**: Calculated as Long-Term Debt divided by Total Equity.
- **ROE**: Return on Equity, calculated as Net Income divided by Total Equity.
- **LogTotalAssets**: The natural logarithm of total assets, serving as a proxy for firm size.

OLS Regression Results						
=====						
Dep. Variable:	vol_std	R-squared:	0.041			
Model:	OLS	Adj. R-squared:	0.041			
Method:	Least Squares	F-statistic:	71.91			
Date:	Tue, 09 Dec 2025	Prob (F-statistic):	2.84e-74			
Time:	21:41:13	Log-Likelihood:	33343.			
No. Observations:	10683	AIC:	-6.667e+04			
Df Residuals:	10677	BIC:	-6.663e+04			
Df Model:	5					
Covariance Type:	HC1					
=====						
	coef	std err	z	P> z	[0.025	0.975]

const	0.0432	0.002	26.908	0.000	0.040	0.046
LogTotalAssets	-0.0010	6.82e-05	-15.219	0.000	-0.001	-0.001
Is_Brown	-0.0002	0.000	-1.122	0.262	-0.001	0.000
Is_High_ESG	0.0003	0.000	1.428	0.153	-0.000	0.001
LeverageRatio1	0.0006	0.000	4.782	0.000	0.000	0.001
ROE	-0.0085	0.001	-11.518	0.000	-0.010	-0.007
=====						
Omnibus:	8515.537	Durbin-Watson:	1.095			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	389593.440			
Skew:	3.471	Prob(JB):	0.00			
Kurtosis:	31.759	Cond. No.	369.			
=====						

The regression results are presented in the table above. The model demonstrates strong joint statistical significance, as evidenced by an F-statistic of 71.91 ($p < 0.01$). The Adjusted R² is 0.041, indicating that the selected variables explain approximately 4.1% of the variance in realized volatility.

The financial control variables behave according to standard economic theory and are all statistically significant at the 1% level ($p < 0.01$):

- **Size Effect (LogTotalAssets):** The coefficient is negative ($\beta = -0.0010$) and highly significant ($t = -15.22$). This confirms that larger firms benefit from diversification and stability, resulting in lower volatility.
- **Profitability (ROE):** Profitability acts as a risk mitigator. A higher Return on Equity is associated with a significant reduction in volatility ($\beta = -0.0085$).
- **Financial Risk (LeverageRatio1):** Financial leverage is positively correlated with risk. The coefficient is positive ($\beta = 0.0006$), implying that for every unit increase in the Long-Term Debt-to-Equity ratio, volatility increases, holding other factors constant.

Regarding the central hypothesis of this study whether "green" allow to reduce volatility, the results from this specification are not statistically significant:

- The coefficient for **Is_High_ESG** is 0.0003 with a p-value of 0.153. We cannot reject the null hypothesis; therefore, achieving an ESG score above 70 does not provide a statistically significant reduction in volatility compared to lower-scoring firms in this sample.
- Similarly, the **Is_Brown** variable ($\beta = -0.0002$) is not statistically significant ($p = 0.262$). This suggests that simply having a Scope 1 intensity above the median does not dictate the level of volatility when controlling for size and leverage.

These findings suggest that market volatility is primarily driven by fundamental financial health (size, leverage, and profitability) rather than environmental labeling. Within this model framework, being classified as "Green" (High ESG) or "Brown" (High Scope 1 intensity) does not have a statistically impact on stock volatility.

This conclusion synthesizes the conflict between your statistical regression (which finds no significance) and your risk modelling (which finds Brown firms riskier in extreme scenarios).

Conclusion Stress Test Part:

The investigation into whether "Green" firms exhibit lower financial stress than "Brown" firms reveals a complex reality where tail risk and fundamental financials diverge. While the initial hypothesis suggested that green firms would inherently display lower volatility, the results across our four testing methodologies present a nuanced answer.

1. Tail Risk vs. Average Volatility

Our stress tests and Monte Carlo simulations highlight a critical distinction between average daily volatility and extreme tail risk.

- **On average**, Green firms do not consistently outperform Brown firms; in the 2008 specific stress test we did, they even had slightly on average, higher risk.
- **In extreme scenarios**, however, Brown firms consistently drive the worst outcomes. The Monte Carlo simulations and the worst-case historical ES metrics confirm that **Brown firms are more susceptible to "fat tail" events**. When the market suffers extreme shocks, high-emission firms experience slightly deeper losses than their green firms.

2. The Importance of Weighting

The sectoral analysis demonstrates that the perception of risk is heavily dependent on the weighting methodology. Capitalization-weighting masks environmental risks by favoring large-cap stability. Conversely, weighting by Scope 1 intensity reveals significant latent risk in the Materials, Utilities, and Industrial sectors. This confirms that financial scale and environmental impact do not always align, and a standard market-cap approach may underprice the risk inherent in high-carbon sectors.

3. Fundamentals Indicators are more Dominant than ESG labels

Finally, the OLS regression provides a necessary reality check. It indicates that volatility is primarily driven by financial fundamentals specifically Size, Leverage, and Profitability (ROE) rather than environmental labeling. The lack of statistical significance for the `Is_High_ESG` and `Is_Brown` variables suggests that, in non-crisis periods, the market prices risk based on the balance sheet rather than the carbon footprint.

IV) Sector-Level Financial Cycle

The objective of this section is to construct a quantitative measure of the financial cycle at the sector level, which will subsequently allow for the analysis of the risk profiles of green and brown firms under comparable market conditions. Any observed relationship between ESG and risk may indeed be entirely driven by the prevailing financial cycle: a sector in an expansion phase mechanically experiences compressed risk premia, whereas during downturns extreme losses and volatility increase, independently of the ESG characteristics of firms. It is therefore crucial to disentangle what reflects the long run structural evolution of sectors from what corresponds to short-run cyclical market dynamics.

Initial investigations were conducted at the daily frequency, notably using the Hodrick–Prescott filter. These attempts proved to be unsatisfactory. Daily data are dominated by excessive noise that prevents the proper identification of meaningful economic dynamics. Moreover, the HP filter produces cycles that are highly sensitive to the choice of the smoothing parameter and often confounds trend and cyclical components, especially near sample endpoints. As a consequence, the extracted components were difficult to interpret economically and unstable over time.

To overcome these limitations, the adopted methodology relies on two major methodological adjustments: aggregation of the data to the monthly frequency and the use of a structural time-series model estimated via the Kalman filter.

Starting from daily individual price data, logarithmic prices are computed and then aggregated by GICS sector and by month. The monthly sector price index is defined as:

$$\text{sector_logprice}_{s,t} = \frac{1}{N_s} \sum_{i \in s} \log(P_{i,t})$$

where $P_{i,t}$ denotes the closing price of firm i belonging to sector s in month t , and N is the number of firms within sector s .

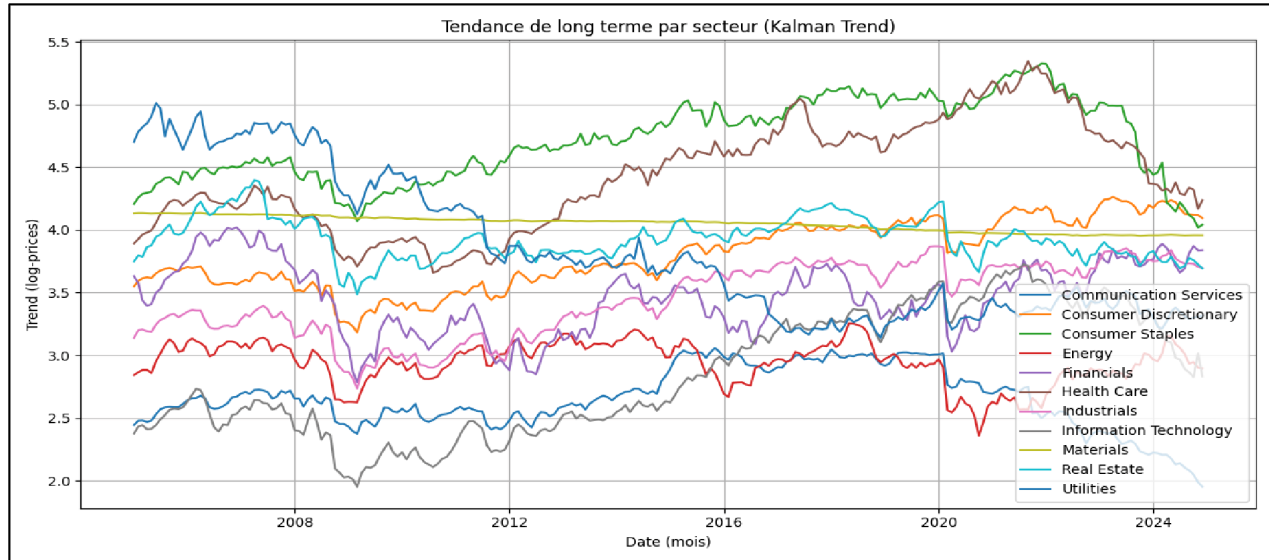
This aggregation provides two key advantages. First, it substantially reduces high frequency noise while preserving the relevant macro-financial information. Second, it aligns the data with the natural time horizon of ESG and risk analysis, which is typically conducted at the monthly or quarterly frequency.

Based on these monthly sector-level series, a state-space unobserved components model is then estimated, of the following form:

$$\forall(s, t), \quad \text{sector_logprice}_{s,t} = \text{trend}_{s,t} + \text{cycle}_{s,t} + \varepsilon_{s,t}.$$

The trend component is modeled as a stochastic local level, capturing the slow-moving and structural evolution of sector valuations. The cycle component is specified as a stochastic harmonic cycle of the AR(2) type, with a fixed period of approximately 30 months, corresponding to the average duration observed for medium-term financial cycles.

This framework produces quasi-sinusoidal cycles whose amplitude and timing are entirely determined by the data through maximum likelihood estimation. The Kalman filter provides smoothed state estimates for each date, which directly yield the corresponding trend and cycle series.



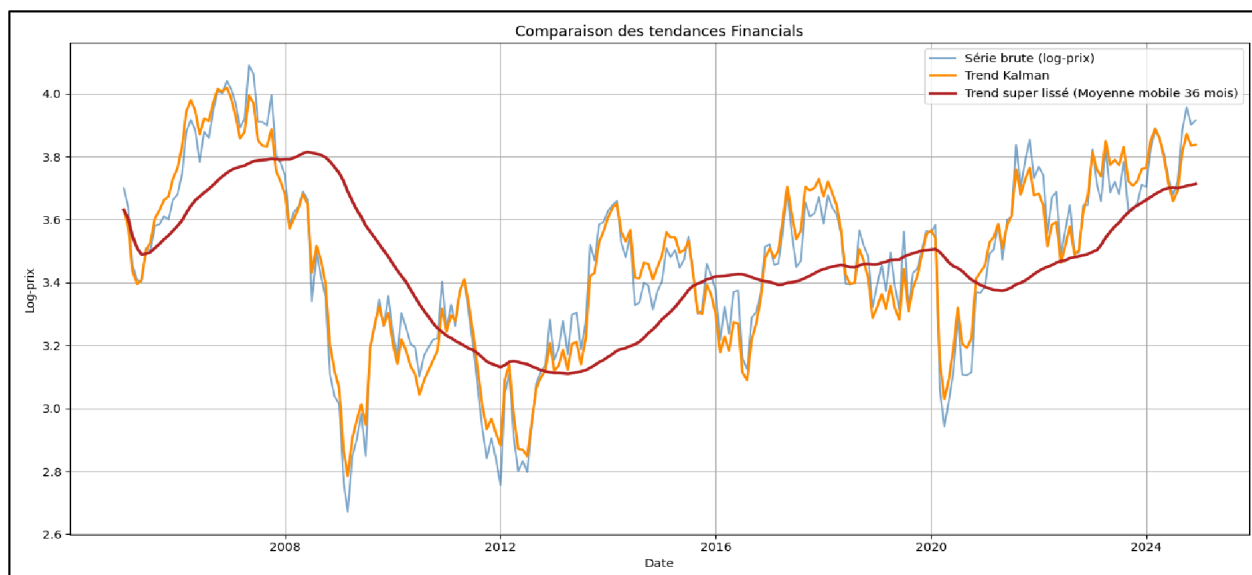
A. Analysis of Sectoral Trends and Cycles

The long-term trends estimated via the Kalman filter highlight markedly different structural trajectories across sectors. Some sectors exhibit an almost continuous revaluation throughout the sample period, particularly Health Care and Consumer Staples, whose dynamics appear steady and resilient. Other sectors display much more uneven or stagnant trajectories, such as Materials or Utilities, where upward and downward phases tend to offset each other over time. Global breaks emerge during major macro-financial shocks, most notably in 2008-2009 during the global financial crisis and again in 2020 with the COVID-19 shock, when all sectoral trends experience a slowdown before gradually readjusting.

To enhance the readability of these structural profiles, the trends extracted from the Kalman filter are subsequently smoothed using a 36-month moving average:

$$\forall(s, t), \quad \text{trend}_{s,t}^{MA} = \frac{1}{36} \sum_{k=0}^{35} \text{trend}_{s,t-k},$$

This long-horizon smoothing removes medium-term fluctuations and clearly highlights the fundamental trajectories of each sector. It allows for a direct comparison of structural regimes: some sectors display a gradual rise in their long-term valuation, while others enter a phase of structural deterioration toward the end of the sample period, particularly within communication services and certain network-based industries.



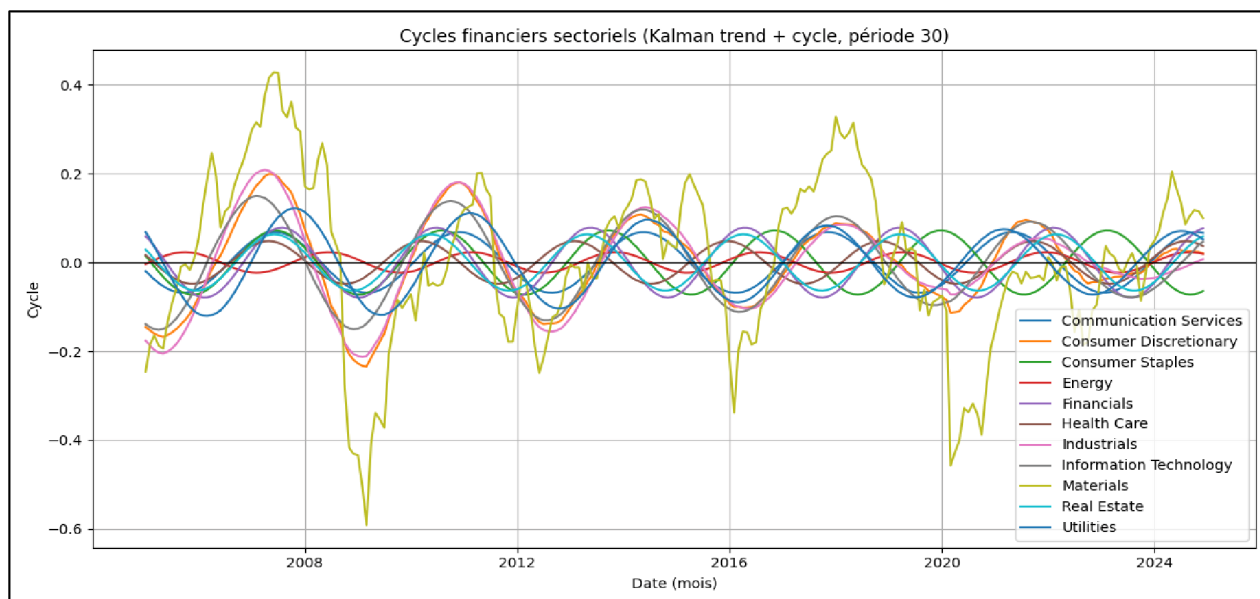
A detailed analysis of the Financials sector fully illustrates the value of this methodology. The raw monthly series remains highly volatile and difficult to interpret. The Kalman trend extracts a much clearer trajectory, revealing the pre-2008 overvaluation, the abrupt collapse during the global financial crisis, and a prolonged phase of incomplete recovery, intermittently disrupted by the 2020 shock. The heavily smoothed trend clearly shows that the losses incurred in 2008 were never fully reversed, which helps explain the sector's persistent fragility despite episodes of apparent recovery.

The sectoral financial cycles, computed as deviations from the trend, provide a direct reading of market regimes.

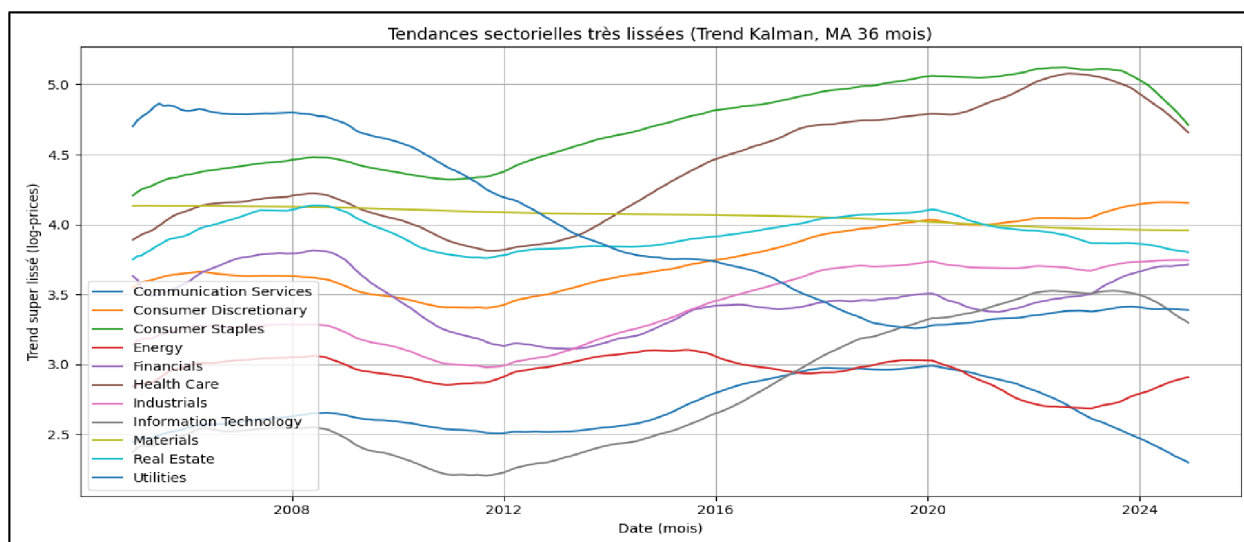
$$\text{cycle}_{s,t} > 0$$

When the sector is in a phase of financial expansion: prices remain persistently above their fundamental trend, risk appetite is elevated, and risk premia are compressed. Conversely, when the sector experiences a phase of contraction or financial stress, characterized by market corrections and increased risk aversion.

$$\text{cycle}_{s,t} < 0$$



The results reveal a very clear synchronization across sectors during major systemic crises, particularly in 2008-2009, in the early 2010s, and most strikingly in 2020, confirming the presence of a common macro-financial cycle. However, the amplitude of these cycles varies substantially across sectors. Energy and Materials display pronounced fluctuations around their trends, reflecting their strong sensitivity to commodity price shocks and global industrial cycles. In contrast, Consumer Staples and Utilities exhibit much weaker cyclical movements, consistent with their defensive nature and low exposure to economic fluctuations.



Overall, this trend-cycle decomposition provides a rigorous and economically coherent interpretation of sector-level market dynamics. It enables the simultaneous identification of each sector's long-term structural trajectory and its precise cyclical position at every point in time. These results constitute a crucial foundation for the remainder of the analysis, as they make it possible to investigate the risk profiles of green and brown firms while explicitly accounting for the market environment in which these risks materialize, thereby avoiding any confusion between cyclical effects, structural trends, and firm-specific characteristics.

a. Econometric Analysis of the ESG – Extreme Risk Relationship

The objective of this section is to examine the relationship between firms' ESG performance and their exposure to extreme risk, measured by the 99% daily Expected Shortfall (ES), while controlling for the market structural dynamics previously identified through the sectoral trend–cycle decomposition obtained via the Kalman filter.

Several model specifications were initially explored: models based on continuous carbon intensity measures (Scopes normalized by sales), models relying on binary green indicators by scope, and specifications incorporating multiple ESG dimensions simultaneously. These preliminary estimations yielded statistically significant effects that were sometimes difficult to interpret jointly, largely due to strong correlations across ESG variables and the large number of parameters involved.

To preserve a clear and economically interpretable framework, the main specification retained is the most parsimonious model, relying exclusively on the Scope 1 green dummy, combined with the sector-level dynamics extracted from the Kalman filter (trend and cycle), as well as sector and year fixed effects. This approach allows for the identification of a directly interpretable ESG effect, while tightly controlling for the macro-financial regime common to firms within the same sector and year

Model Specification :

The estimated model is specified as follows:

$$ES_{i,t} = \alpha + \theta \text{Green_scope1}_i + \phi_1 \text{trend}_{s(i),t} + \phi_2 \text{cycle}_{s(i),t} + \gamma_{s(i)} + \delta_{\text{year}(t)} + u_{i,t}$$

- ES is the 99% daily Expected Shortfall of firm (i) at time (t),
- Greenscope1 is a binary indicator equal to 1 if the firm belongs to the *green* group under Scope 1, and 0 otherwise,
- trend and cycle denote respectively the long-term trend and the cyclical component of sector (s(i)), as obtained from the Kalman filter,
- γ represents sector fixed effects
- δ denotes year fixed effects
- And u is the idiosyncratic error term

The model is estimated by Ordinary Least Squares using approximately 460,000 daily observations, with HC3 heteroskedasticity-robust standard errors to account for the pronounced heteroskedasticity characteristic of financial risk data.

Presentation of Results :

The estimated coefficients from the simplified specification are reported in the main regression table:

TABLE 1 – Modèle 3 simplifié : ES sur facteur ESG et dynamique sectorielle

	Coefficient	Statistique <i>z</i>	<i>p</i> -valeur
Intercept	0.0448***	90.045	0.000
Green_scope1	0.0018***	9.425	0.000
trend	-0.0064***	-36.044	0.000
cycle	-0.0363***	-65.819	0.000
Effets fixes secteur	Oui (C(GICS_Sector))		
Effets fixes année	Oui (C(year))		
R^2	0.112		
Nombre d'observations	464 494		

Erreurs standards robustes (HC3).

*** $p < 0,01$; ** $p < 0,05$; * $p < 0,10$.

The first results concern the sectoral and time fixed effects. Most sector coefficients are statistically significant, indicating substantial structural differences in extreme risk levels across sectors. Similarly, the annual fixed effects reach particularly high magnitudes during periods of systemic stress such as 2008-2009, 2011, 2020, and 2022, confirming the strong dependence of Expected Shortfall on global macrofinancial regimes. These variables primarily serve as control factors, allowing firms to be compared within identical sectoral and yearly conditions. The key variables of interest are the ESG dummy (*Green_scope1*) and the sectoral dynamic components (*trend* and *cycle*).

The coefficient on *Green_scope1* is positive and statistically significant at the 1% level ($\beta \approx 0.0018$). All else being equal—that is, after controlling for global market conditions and sectoral structure—firms classified as *green* under Scope 1 exhibit a slightly higher average level of Expected Shortfall compared to non-green firms. Although the magnitude of the effect remains modest, it is statistically robust. From an economic perspective, this result suggests that membership in the environmentally most virtuous group under Scope 1 does not automatically translate into reduced exposure to extreme downside risk. This pattern can be interpreted as reflecting transitional dynamics associated with decarbonisation processes: heavy investment requirements, regulatory uncertainty, and competitive pressures may temporarily increase firms' sensitivity to adverse shocks, even as they improve their ESG outlook over the longer term. The macro-sectoral components play a central role in determining extreme risk exposure. The coefficient on the sectoral trend is negative and highly significant ($\beta \approx -0.0064$). When sector price levels stand above their long-term trend, Expected Shortfall decreases, indicating a compression of tail risk during buoyant market conditions. Conversely, episodes of structural sectoral weakness are associated with heightened extreme risk. Similarly, the cycle variable exhibits a negative coefficient ($\beta \approx -0.0363$). This indicates that cyclical deviations from the trend—particularly during turning points of the financial cycle—substantially amplify tail risk, confirming the empirically procyclical behaviour of Expected Shortfall.

Model Fit and Interpretation

The model achieves an R^2 of approximately 11%, a level that is relatively modest but entirely standard in financial regressions involving daily measures of extreme risk, which are inherently noisy and influenced by numerous unobserved determinants. The goal of the model is not to provide a comprehensive forecasting tool, but rather to uncover conditional structural relationships between ESG profiles, sectoral regimes, and tail risk exposure.

Within this framework, three robust conclusions emerge:

- Macro-sectoral dynamics (*trend* and *cycle*) constitute the dominant drivers of Expected Shortfall.
- The Scope 1 ESG effect, while quantitatively limited, is statistically significant and positive.
- Membership in the *green* group does not mechanically imply reduced short-term exposure to extreme risk, suggesting that the energy transition may entail temporary financial costs that manifest in tail-risk measures.

Conclusion :

This research aimed to determine whether Green firms truly offer a safer investment alternative compared to their Brown counterparts. By rebuilding the J.P. Morgan RiskMetrics framework with Student-t assumptions, we were able to model financial returns with greater accuracy, particularly regarding extreme tail risks. Our findings provide a nuanced answer to the central question, revealing that the relationship between ESG and risk is far more complex than a simple binary distinction.

First, our marginal risk analysis reveals a clear difference in extreme scenarios. While average daily volatility often shows little difference between the two groups, Brown firms exhibit significantly heavier tails and higher Expected Shortfall during stress periods. This confirms that high-emission companies are indeed more prone to severe crashes. However, our Copula-based dependence analysis highlights a persistent systemic link. Green and Brown portfolios remain strongly correlated, especially during market downturns. This implies that while Green assets are individually safer regarding tail risks, they do not provide a complete shield against systemic shocks, as they tend to move in sympathy with the broader market during crises.

Furthermore, our decomposition of the financial cycle using the Kalman filter demonstrated that sector-level dynamics are the primary drivers of risk. When controlling for these cycles and fundamental financial metrics such as size and leverage, the ESG label itself loses much of its explanatory power for daily volatility. Our regression models even suggest that the transition to a low-carbon model may induce temporary uncertainty. In conclusion, ESG criteria are a valuable tool for mitigating extreme tail risk, but they are not a substitute for fundamental risk management. A firm's environmental score changes the shape of its risk profile rather than eliminating volatility entirely, meaning investors must still account for global financial cycles and traditional balance sheet fundamentals.